5(a)(i)	Φ is work function energy and refers to the minimum amount of energy needed to liberate/free an electron from a metal (surface).	[1]
(ii)	E_K is the maximum/highest possible kinetic energy of an emitted electron.	[1]
(b)(i)	By photoelectric effect equation, Using $hf = \phi + \frac{1}{2}mv_{max}^2$ $(6.63 \times 10^{-34} \times 3 \times 10^{8})/ 300 \times 10^{-9} = (2.03)(1.6 \times 10^{-19}) + eV_s$ $eV_s = 3.38 \times 10^{-19}$ J = 2.11 eV $V_s = 2.11$ V	[1] [1]
(ii)	$\text{intensity} = \dfrac{power}{area} = \left(\dfrac{n}{t}\right)_p \dfrac{hf}{A}$ $0.500 = \left(\dfrac{n}{t}\right)_p \dfrac{\left[\dfrac{(6.63\times10^{-34})(3\times10^{8})}{300\times10^{-9}}\right]}{4.5\times10^{-5}}$ $\left(\dfrac{n}{t}\right)_p = 3.39\times10^{13}$ $\left(\dfrac{n}{t}\right)_e = 1.7\times10^{13}$ $i = \left(\dfrac{n}{t}\right)_e e = (1.7\times10^{13})(1.6\times10^{-19}) = 2.72\times10^{-6}$ A	 [1] [1] [1]
(c)	Some of the incident photons were reflected from the surface of the metal and therefore not all the photons incident was absorbed. OR The second reason would be that on the way out to the metal surface, an electron may lose its (kinetic) energy to ions and other electrons it encounters along the way. This energy loss may prevent some electrons from overcoming the work function.	[1] [1]
(d)(i)	From the graph, $\lambda_{min} = 1.2 \times 10^{-11}$ m Hence max energy of photons $= h c / \lambda_{min}$ $= 6.63 \times 10^{-34} \times 3 \times 10^{8} / 1.2 \times 10^{-11}$ $= 1.66 \times 10^{-14}$ J	[1] [1]
(ii)	By conservation of energy when electrons are accelerated by the electric field, KE gain = Electrostatic Potential energy loss i.e. KE = (e ΔV) When the accelerated electron makes a collision with the target atoms such that its KE is totally converted to the energy of an X-ray photon, then KE = 1.66×10^{-14} J (fr (i)) Hence (e ΔV) = 1.66×10^{-14} i.e. $\Delta V = 1.66 \times 10^{-14} / 1.6 \times 10^{-19}$ = 103.5 kV ≈ 104×10^{3} V	 [1] [1]

(iii)	<u>same starting point for wavelength</u>; curve follows original but is <u>higher</u> than given curve (as more electrons per unit time are produced and the number of photons per unit time will be more)	[1]
(iv)	If 600 W (of heat generated) is 99.5 % of total power input, then total power is (100/99.5) x 600 = 603 W which is the total energy of electrons arriving per second at the target. Hence number of incident electrons per second x energy of each electron = 603 $n \times 1.66 \times 10^{-14} = 603$ $n = 3.63 \times 10^{16}\ s^{-1}$	[1] [1]